**Series:** An *series* is an infinite sum of terms

adding infinitely many terms together

$$\sum_{i=1}^{\infty} a_i = a_1 + a_2 + \dots$$

stopping index

starting index

```
for(i = 1){
  sum = sum + a[i]
}
```

which we understand as the limit of a sequence of partial sums $\{s_n\}$ given by

$$s_1 = a_1, \qquad s_2 = a_1 + a_2, \quad \dots \quad s_n = \sum_{i=1}^{n} a_i,$$

meaning (so long as the limit exists),

$$\sum_{i=1}^{\infty} a_i = \lim_{n\to\infty} s_n.$$

We say $\sum_{i=1}^{\infty} a_i$ is:

- *Convergent* if $\{s_n\}$ is convergent.
- *Divergent* if $\{s_n\}$ is divergent.

**Example:** Determine the behavior of

$$\sum_{i=1}^{\infty}(5i+3).$$

Divergent series

$s_1 = \sum_{i=1}^{1}(5i+3) = 5(1)+3 = 8$

$s_2 = \sum_{i=1}^{2}(5i+3) = 8+(5(2)+3) = 21$

$s_3 = \sum_{i=1}^{3}(5i+3) = 8+21+(5(3)+3) = 39$

$S_n = \infty$

↳ Diverges

It will converge if what's being summed starts to become negligible?

**Example:** Find

j = index

$$\sum_{j=1}^{\infty}\left(\frac{1}{j} - \frac{1}{j+1}\right) = \lim_{n\to\infty}\left(1 - \frac{1}{n+1}\right) = 1$$

convergent series

← converges

$S_1 = 1 - \frac{1}{2}$ $\quad S_2 = \left(1 - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \frac{1}{3}\right)$ $\quad S_3 = \left(1 - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}\right) + \left(\cancel{\frac{1}{3}} - \frac{1}{4}\right)$

⋮

$$S_n = \underbrace{1 - \frac{1}{n+1}}_{} = \left\{1 - \frac{1}{n+1}\right\}$$

you cannot take n and plug it in for all the js

you end up getting a sequence

We interpret $\sum_{i=1}^{\infty} a_i = \lim_{n \to \infty} S_n$ where $\underbrace{S_n = \sum_{i=1}^{n} a_i}_{\text{partial sum from } a_1 \text{ to } a_n?}$

**Example:** Find

$$\sum_{m=1}^{\infty} \frac{7}{m^2 + 3m + 2}.$$

Even if it's positive it doesn't mean it will diverge

*Hint:* Use a partial fraction decomposition.

$$\sum_{m=1}^{\infty} \frac{7}{m+1} - \frac{7}{m+2}$$

$$\frac{7}{(m+1)(m+2)} = \frac{A}{m+1} + \frac{B}{m+2}$$

$$= \frac{7}{m+1} - \frac{7}{m+2}$$

$7 = A(m+2) + B(m+1)$

When $m = -2 \to 7 = B(-1) \to B = -7$

When $m = -1 \to 7 = A(1) \to A = 7$

$$S_1 = \left(\frac{7}{2} - \frac{7}{3}\right)$$

$$S_2 = \left(\frac{7}{2} - \frac{7}{3}\right) + \left(\frac{7}{3} - \frac{7}{4}\right)$$

$$S_3 = \left(\frac{7}{2} \cancel{- \frac{7}{3}}\right) + \left(\cancel{\frac{7}{3}} \cancel{- \frac{7}{4}}\right) + \left(\cancel{\frac{7}{4}} - \frac{7}{5}\right)$$

$s = 3 \to 5 = s+2$

$$S_n = \left(\frac{7}{2} - \frac{7}{n+2}\right)$$

**Geometric Series:** For nonzero real numbers $a, r$, consider

$$\sum_{i=1}^{\infty} ar^i.$$

where $a, r$ are real numbers

$$\sum_{i=0}^{\infty} ar^i$$

After subtracting

$$S_n = a + \cancel{ar} + \cancel{\cdots} + \cancel{ar^n}$$

$$rS_n = \cancel{ar} + \cancel{\cdots} + \cancel{ar^n} + ar^{n+1}$$

$$S_n - rS_n = a - ar^{n+1}$$

$$S_n(1-r) \to S_n = \frac{a - ar^{n+1}}{1-r}$$

$$S_n = \frac{a - ar^{n+1}}{1-r}$$

$$\sum_{i=0}^{\infty} ar^i = \lim_{n \to \infty} S_n = \lim_{n \to \infty} \frac{a - ar^{n+1}}{1-r}$$

when $|r| < 1 \to \frac{a}{1-r}$

when $|r| > 1 \to$ Diverges

$$\sum_{i=0}^{\infty} ar^i = ar^0 + \sum_{i=1}^{\infty} ar^i$$

$$= a + \sum_{i=1}^{\infty} ar^i$$

So $\sum_{i=1}^{\infty} ar^i = \frac{a}{1-r} - a$?

**Example:** Determine whether

$$\sum_{i=0}^{\infty} 3^{2n} 4^{n-1} = \sum_{i=0}^{\infty} \frac{9^n}{4^n} \cdot 4 \qquad \sum_{i=0}^{\infty} ar^i$$

$4 \times 4$

$$= \sum_{i=0}^{\infty} 4\left(\frac{9}{4}\right)^n \to a = 4, \quad r = 9/4 \to \left|\frac{9}{4}\right| > 1 \to \text{Diverges}$$

converges or diverges.

$$\sum_{i=0}^{\infty} 3^{-2n} 4^{n-2} = \sum_{i=0}^{\infty} \frac{1}{4}\left(\frac{4}{9}\right)^n = \frac{1/4}{1 - 4/9}$$

$a = \frac{1}{4}$

$r = \frac{4}{9} \to |r| < 1$

**Series Properties:** Suppose

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} b_n$$

both converge. Then for any real number $c$,

$$\sum_{n=1}^{\infty} ca_n = c \sum_{n=1}^{\infty} a_n.$$

$$\sum_{n=1}^{\infty} a_n \pm b_n = \sum_{n=1}^{\infty} a_n \pm \sum_{n=1}^{\infty} b_n.$$

Further, for any $k \geq 1$,

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=k}^{\infty} a_n$$

either both converge or both diverge.

k could = 2, 3, 50 or

**Example:** Find

$$\sum_{n=1}^{\infty} \frac{4}{n(n+1)} - \frac{1}{3^n}.$$

$$Ex - \sum_{i=1}^{\infty} \frac{4}{i(i+2)} - \frac{1}{3^i}$$

$$= \sum_{i=1}^{\infty} \frac{4}{i(i+1)} - \sum_{i=1}^{\infty} \frac{1}{3^i}$$

$$= 4 \sum_{i=1}^{\infty} \frac{1}{i(i+1)} \qquad = \sum_{i=1}^{\infty} \left(\frac{1}{3}\right)^i$$

if one is divergent then the whole thing is

Getting a geometric series since we know what that looks like

$$\sum_{i=0}^{\infty} \left(\frac{1}{3}\right)^i = \frac{1}{1 - 1/3} = \frac{3}{2}$$

geometric sequence - know to be true

or

$$\sum_{i=1-1}^{\infty} \left(\frac{1}{3}\right)^{i+1} = \sum_{i=0}^{\infty} \frac{1}{3}\left(\frac{1}{3}\right)^i$$

why?

$$= \left(\frac{1}{3}\right)^0 + \underbrace{\sum_{i=1}^{\infty} \left(\frac{1}{3}\right)^i}_{\text{want to find}}$$

$$\text{So } \sum_{i=1}^{\infty} \left(\frac{1}{3}\right)^i = \frac{3}{2} - 1 = \frac{1}{2}$$

Other one

$$4 \sum_{i=1}^{\infty} \frac{1}{i(i+1)}$$

$$\frac{1}{i(i+1)} = \frac{A}{i} + \frac{B}{(i+1)} = \frac{1}{i} - \frac{1}{i+1}$$

$$1 = A(i+1) + Bi$$

$$A = 1 \quad B = -1$$

$$= 4 \sum_{i=1}^{\infty} \frac{1}{i} - \frac{1}{i+1}$$

**Harmonic Series:** The *harmonic series*

$$\sum_{i=1}^{\infty} \frac{1}{i} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

diverges. However, the *alternating harmonic series*

$$\sum_{i=1}^{\infty} (-1)^{i+1} \frac{1}{i} = 1 - \frac{1}{2} + \frac{1}{3} - \dots$$

converges.

**Example:** Show that

$$\sum_{j=1}^{\infty} \frac{1}{j+3}$$

diverges.

$$= \sum_{j=4}^{\infty} \frac{1}{j} \quad \text{Diverges}$$

↑ Harmonic Series starting at 4

Since at $j=4$ it diverges it will also diverge at any other starting index

(this is also true for convergence)

**Theorem:** If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

**Divergence Test:** If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

**Example:** Show that

$$\sum_{l=1}^{\infty} \frac{l^2}{4l^2 + 7l + 3}$$

diverges.

**Example:** Show that

$$\sum_{h=1}^{\infty} \frac{\cos^2(h)}{10}$$

diverges.